CCD-TBLCA: A Land Cover Continuous Change Detection Method Fusing Transformer and Bi-Directional LSTM via Cross-Attention

Shanlin Gao, Qizhi Zhuang, Yi Song, Feng Zhu, Liang Cheng, and Manchun Li

***Abstract*—Continuous land cover changes in remote sensing observations can provide crucial data support for evaluating urban spatial planning and sustainable development, but their acquisition remains challenging because it must overcome missing observations caused by clouds and cloud shadows while accurately characterizing the spatiotemporal dynamics of land cover semantics. Therefore, we proposed a continuous change detection method fusing transformer and bi-directional long short-term memory (LSTM) via cross-attention (CCD-TBLCA) in land cover using Sentinel-2 imagery. Specifically, it employed a dual-branch encoding design for spectral time-series data, enabling deep information fusion of long-term dependencies and local dynamics. First, we used the optical temporal information from raw irregular time series, achieving monthly mapping from spectral anomalies to land cover change events based on CCD-TBLCA. Subsequently, we proposed a breakpoint detection criterion that enforced semantic consistency before and after changes, thereby forming a more comprehensive framework for evaluating breakpoint detection alongside spatial existence and temporal accuracy. Across seven urban regions, CCD-TBLCA outperformed existing methods in both temporal classification and breakpoint detection, with semantic-consistent breakpoint detection accuracy generally improving by over 15%. In spatial mapping, CCD-TBLCA achieved F1 accuracy above 0.9 for most land cover types, surpassing existing products. CCD-TBLCA's robustness to sequence updates and irregular time series had also been validated. The results show that CCD-TBLCA provided in this study can accurately describe continuous spatiotemporal land cover changes and help to understand how human activities reshape land cover patterns.**

***Index Terms*—Bi-directional long short-term memory (BiLSTM), continuous change detection, land cover, semantic consistency, Sentinel-2, transformer.**

I. Introduction

DURING rapid urban development, complex human activities continually reshape land cover patterns [1], [2]. Driven by regional economies and land use policies, urban expansion and industrial growth often cause ecological degradation and encroachment on agricultural resources [3], [4]. The dynamics of built-up areas and other land cover types provide traceable evidence for assessing urban planning, ecological protection, and cropland security [5], [6], [7]. Continuous, timely, and accurate land cover mapping and change detection are therefore essential for characterizing human-driven spatiotemporal dynamics of the land surface and for analyzing the balance among urban development, ecological environment, and agricultural resources, and they have become important research topics [8], [9], [10].

Improvements in satellite revisit frequency have enabled dense, high-frequency Earth observation. In parallel, continuous land cover change detection has evolved from classical spectral time-series models to deep-learning-based frameworks [11], [12]. Classical models interpret time-series characteristics through matching [13], thresholding [14], segmentation [15], and decomposition [10], [16]. Matching relies on the similarity between observations and templates, while thresholding relies on the deviation between observations and a threshold. Segmentation divides the time series into several line segments to obtain breakpoints. Decomposition models components such as trend and seasonality, then detects changes through abnormal deviations. Although classical models have been widely used [17], [18], algorithms such as BFAST and LandTrendr process only single-band data and simultaneously distinguish multiple categories [19]. Continuous change detection and classification (CCDC) handles multiband data, but may miss early changes [20]. Their performance is also sensitive to hyperparameters and requires tuning for each application.

Deep learning has introduced convolutional neural networks (CNNs), recurrent neural networks (RNNs), and transformers into land cover mapping and change detection based on multisource remote sensing imagery, particularly for time-series data [21], [22], [23], [24], [25], [26], [27]. However, cloud cover creates gaps and spectral noise, leading to missing observations [28]. 1-D CNNs, such as temporal semantic segmentation change detection (TSSCD) in a fully convolutional network (FCN) form, cannot directly handle missing observations and require sequence interpolation [29], [30]. Even RNNs or transformer-based methods, which tolerate missing values, still use interpolated sequences [31]. However, interpolation may introduce temporal errors in detecting abrupt events [32], [33], [34]. Gating and self-attention are powerful in natural language processing and have been widely applied

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to remote sensing time series [35], [36], [37], [38]. However, RNNs struggle with long-term dependencies, while the transformer relies on positional encoding with limited effectiveness to capture local dynamics [39].

Most studies adopt a sequence-to-one setting that predicts current land cover from historical sequences [36], [37], whereas sequence-to-sequence design that traces land cover dynamics receives less attention [40]. Moreover, although classical and deep-learning-based methods continue to advance, most evaluations still focus on breakpoints' spatial existence and temporal accuracy [10], [30], [41]. They assess whether breakpoints occur and whether their timing is accurate, without considering the land cover semantic information before and after changes. This omission may cause the model's outputs to deviate from the real world.

To address quality deficiencies in remote sensing time series and preserve accurate semantic information during land cover changes, we proposed a continuous change detection method fusing transformer and a bi-directional long short-term memory (LSTM) via cross-attention (CCD-TBLCA), a continuous change detection method that fully leveraged spectral temporal information. CCD-TBLCA deeply fused transformer and bi-directional LSTM (BiLSTM) through cross-attention (CA) to capture long-term dependencies across years and seasons and subtle changes in local dynamics. It achieved end-to-end land cover temporal classification in a sequence-to-sequence setting while maintaining robustness to irregular time series, thereby identifying continuous breakpoints. We further introduced a breakpoint detection metric that enforced semantic consistency before and after changes, complementing the evaluation framework alongside spatial existence and temporal accuracy. On monthly change detection datasets and spatial mapping products, CCD-TBLCA achieved optimal spatial classification and breakpoint detection results. The main contributions are as follows.

1) A dual-branch temporal characterization and deep fusion framework. Symmetric CA was used to achieve complementary fusion of time-series features from transformer and BiLSTM, which was oriented to obtain a more accurate end-to-end sequence modeling capability for continuous changes in multiple land cover types.
2) Robust modeling mechanism for missing observations in remote sensing time series. To address the invalid observations caused by clouds and cloud shadows, we introduced a strategy to extract clear observations by masking the missing values, so that the model can achieve temporal classification directly on the raw irregular time series.
3) Use semantic-consistent breakpoint detection metrics to construct a more complete evaluation system. The semantic consistency metrics were closer to the real land cover change process than one-sided spatial existence and temporal accuracy, and could reflect the modeling accuracy of continuous change detection methods for the real world.
4) Validating model accuracy in cities with different geographic environments. We constructed a monthly-scale land cover continuous change detection dataset in Kunming with a time span of 90 months, and combined it with the existing dataset to construct a validation system including seven geographically distinct cities, in order to systematically evaluate the ability of CCD-TBLCA to detect continuous changes under different climatic, topographic, and land cover patterns.

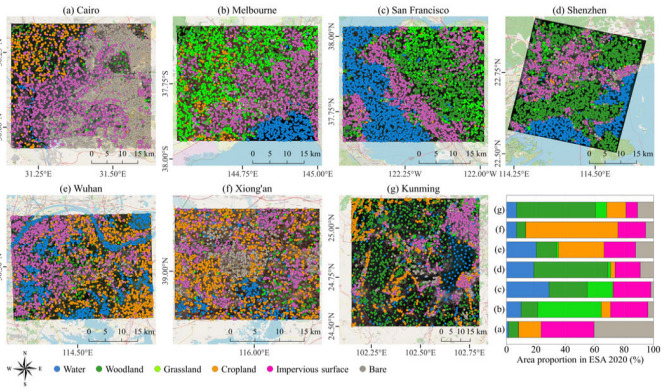


Fig. 1. Spatial locations of samples for different land cover types within each city, along with the proportion of each type's area calculated based on European Space Agency (ESA) WorldCover 2020. (a) Cairo. (b) Melbourne. (c) San Francisco. (d) Shenzhen. (e) Wuhan. (f) Xiong'an. (g) Kunming.

II. Study Area and Data

A. Time-Series Datasets

He et al. [30] constructed a monthly Sentinel-2 L2A composite dataset for pixel-level continuous land cover change detection using samples of different land cover types from multiple cities worldwide. As shown in Fig. 1(a)–(f) and summarized in Table I, the spatial extent of this publicly available dataset includes six cities, including Cairo, Melbourne, San Francisco, Shenzhen, Wuhan, and Xiong'an New Area, and covers six major land cover types, including water, woodland, grassland, cropland, impervious surface (IS), and bare land, with the time span of 48 months (hereafter referred to as the TSSCD-48 dataset).

In the TSSCD-48 dataset, different cities have large differences in climate, topography, and land cover types. Except for Shenzhen and San Francisco, which have a small amount of hilly terrain, the topography of the other cities is relatively flat. In addition, the dominant and missing types of land cover vary. For example, Cairo is dominated by impervious surface and bare land, with a very low percentage of water and grassland; Melbourne is dominated by grassland; Shenzhen is dominated by woodland; Wuhan has minimal grassland, while San Francisco exhibits low proportions of cropland and bare land, with other land cover types being relatively balanced between these two cities; and Xiong'an New Area is dominated by cropland. Finally, the pattern of land cover change varies across cities, with Cairo, Shenzhen, Wuhan, and Xiong'an New Area dominated by urban expansion, Melbourne mainly by vegetation conversion, and San Francisco with no significant change.

To evaluate CCD-TBLCA under diverse geographic conditions, a Kunming dataset with low latitude and high elevation, which was the seventh city, was added. In Fig. 2, the study area

TABLE I
TEMPORAL RANGE, COVERAGE AREA, AND NUMBER OF SAMPLES ACROSS ALL CITIES

Cities	Temporal range	Coverage area/km^2	Samples (stable/changing)
Cairo	Jan 2018 – Dec 2021	2,420	3,828/844
Melbourne	Jan 2019 - Dec 2022	2,213	5,209/582
San Francisco	Jan 2019 - Dec 2022	2,210	5,521/396
Shenzhen	Dec 2017 - Nov 2021	1,422	4,736/1,103
Wuhan	Dec 2018 - Nov 2022	1,826	5,038/1,084
Xiong'an	Jan 2019 - Dec 2022	1,086	3,257/1,296
Kunming	Jan 2018 – Jun 2025	4,800	2,894/827

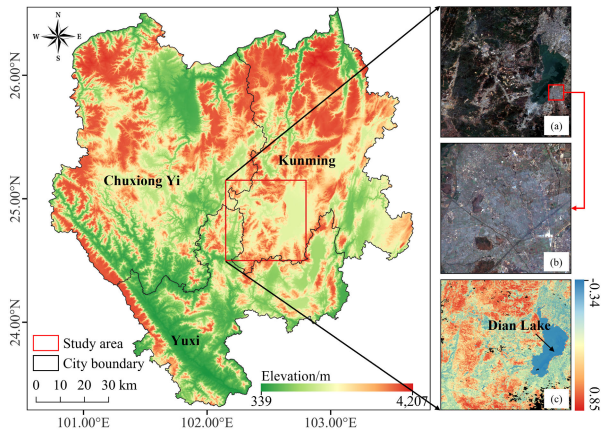


Fig. 2. Location and basic geographic information of the Kunming study area. The red box in (a) highlights the greenhouse cropland. (a) Sentinel-2: 2025-05. (b) Greenhouse cropland. (c) NDVI.

lies at the junction of Kunming, Chuxiong Yi Autonomous Prefecture, and Yuxi. It has an average elevation exceeding 2000 m, covers approximately 4800 km^2, and experiences a monsoon-affected subtropical highland climate with dry and wet seasons. Precipitation concentrates from May to October, leading to frequent cloud cover that complicates long-term monitoring. This region also contains unique greenhouse cropland [Fig. 2(b)] with normalized difference vegetation index (NDVI) signatures distinct from open-air vegetation [Fig. 2(c)] and has recently undergone rapid land cover changes due to mining and urban development. According to Fig. 1, the Kunming dataset is dominated by woodland, with other land cover types exhibiting relatively balanced proportions. The addition of the Kunming dataset introduces challenges such as low latitude, high elevation, and high cloudiness for land cover continuous change detection. The TSSCD-48 dataset shares consistent task definitions and land cover types with the Kunming dataset. However, the differentiation of land cover spatial distribution and change patterns of different cities can provide a reliable and necessary test for evaluating the performance of the CCD-TBLCA in different geographic environments.

Sentinel-2 L1C [i.e., top-of-atmosphere reflectance (TOA)] was used to construct pixel-level time series. On Google Earth Engine (GEE), we generated monthly median composites and resampled each band to 10 m. Clouds and cloud shadows were removed using the cs_cdf band from Cloud Score+ [42]. Pixels with values below 0.65 were masked and filled with −9999. We selected 3721 samples at a ratio of approximately 3:1 between stable and changing samples. Labels were assigned according to land cover in January 2018. Sample locations are shown in Fig. 1(g), and additional details are provided in Table I. Six categories and a transition matrix driven by changing samples in Kunming are shown in Fig. 3. Cropland includes open-air and greenhouse forms. Most transitions occur from woodland to bare land and from bare to impervious surface. Models based on supervised learning paradigms heavily depend on the scale and quality of training datasets [43], [44]. Given that this study aimed to detect all land cover type changes occurring within the time series, accurate annotation of land cover types at each timestep is essential, achieved through high-resolution historical Google Earth imagery and monthly composite Sentinel-2 time-series images. Specifically, we inspected the Sentinel-2 true-color and false-color image thumbnails surrounding each sample to annotate the moment of land cover type changes and the categories before and after the change. To ensure dataset quality, all samples underwent land cover breakpoint annotation by three professional data annotators. Finally, samples with inconsistent breakpoints were discarded. Both the TSSCD-48 and Kunming datasets were split into training, validation, and test sets using a 6:2:2 ratio.

B. Spatial Classification Datasets

To benchmark the spatial mapping performance, four existing products were selected for comparison: China land cover dataset (CLCD) [45], ESA WorldCover [46], [47], Dynamic World [8], and Esri Land Cover [48]. CLCD provides annual land cover maps in China using Landsat. ESA WorldCover offers global maps in 2020 and 2021 derived from Sentinel-1 and Sentinel-2. Dynamic World is a near-real-time global 10 m land cover product based on Sentinel-2 that provides classification probabilities. Esri Land Cover provides annual global 10 m maps based on Sentinel-2. The mapping relationship from these products to the six categories in this study is shown in Table II.

III. METHODS

To capture diverse, multistage changes, we proposed a temporal classification and continuous change detection method suitable for multiple land cover types, as illustrated in Fig. 4. First, we produced monthly composites and constructed each sample's spectral time series. Subsequently, we fused a

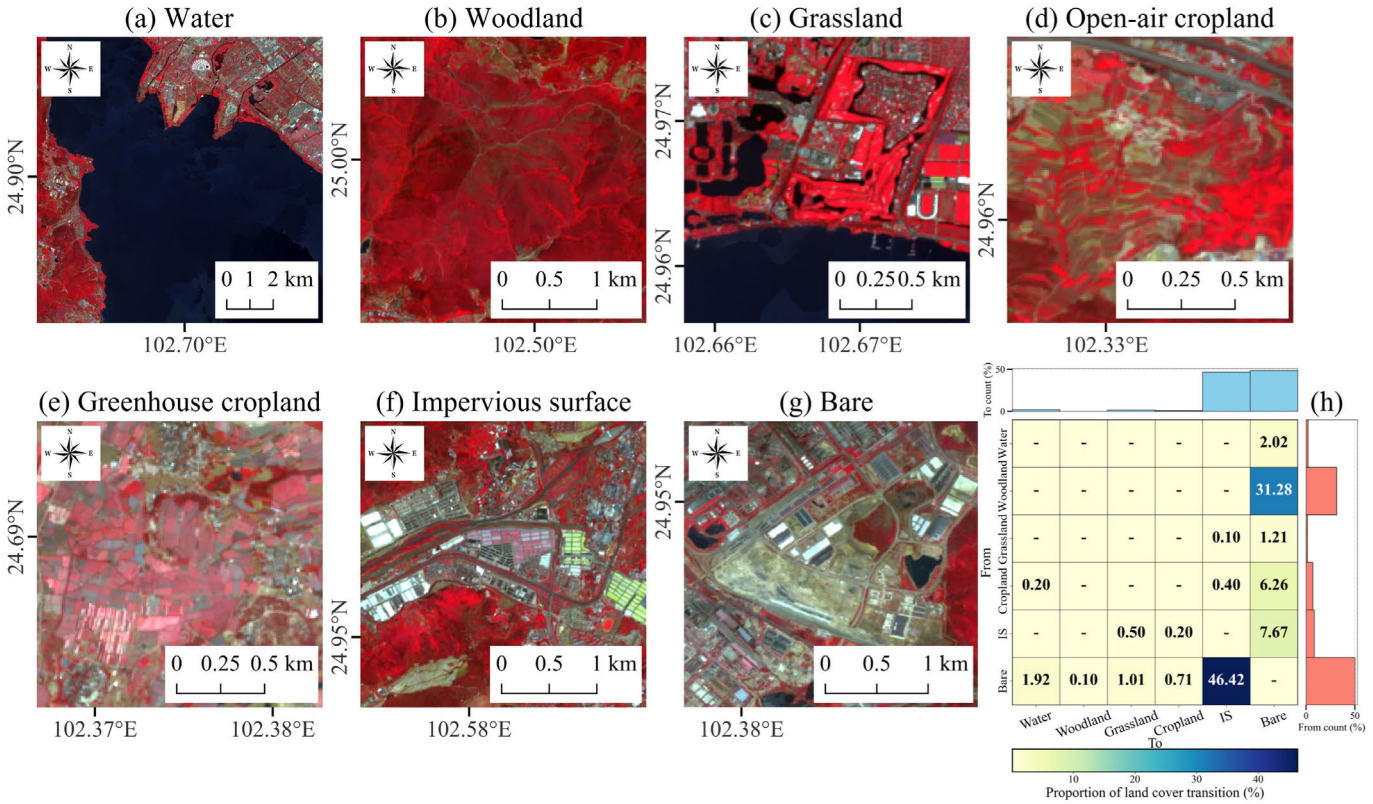


Fig. 3. Six land cover types in Kunming and a transition matrix driven by changing samples. (a) Water. (b) Woodland. (c) Grassland. (d) Open-air cropland. (e) Greenhouse cropland. (f) Impervious surface. (g) Bare land. (h) Land cover transition matrix driven by changing samples.

transformer and BiLSTM to process time series, capturing long-term dependencies and local dynamics. CA achieved deep information fusion between them. Finally, supported by a comprehensive evaluation framework and existing products, we assessed CCD-TBLCA's accuracy in spatial mapping and change detection.

A. Construction of CCD-TBLCA

1) Inputs: Ten raw bands from Sentinel-2 L1C were used as inputs. Each sample's input matrix is $X_{in} \in \mathbb{R}^{L\times C}$, where L is the time-series length and C is the number of bands. For a given timestep t, the inputs are defined in (1). In Fig. 5(c), the inputs pass through a global attention mechanism (GAM) channel attention with residual connection [49] to enhance spectral information

$$X_t^{in} = \left\{x_t^{B2}, x_t^{B3}, x_t^{B4}, x_t^{B5}, x_t^{B6}, x_t^{B7}, x_t^{B8}, x_t^{B8A}, x_t^{B11}, x_t^{B12}\right\}. \quad (1)$$

2) Encoder: The encoder adopted a parallel transformer and BiLSTM branches. For sequence-to-sequence land cover classification, we used an encoder-only transformer with masking so that missing observations did not act as Key. In Fig. 6(a), after masking, inputs are passed through the embedding layer and positional encoding (PE) before entering the transformer block, which computes attention over valid timesteps. Masked timesteps in Query are excluded from loss calculation, thus not affecting gradient propagation. $H_t^T \in \mathbb{R}^d$ is the context feature output at time step t. The encoding information $F_T \in \mathbb{R}^{L\times d}$ is obtained by concatenating allattention (MHA) combines all attention heads H_t^T.

Self-attention (SA) is computed in (2), and multihead attention (MHA) combines all attention heads in (3). Here, h denotes the number of attention heads, $head_i$ represents each head's outputs, and W^O is a learnable matrix. In (4), for masked timesteps, the masking mechanism excludes missing observations from affecting the outputs. Here, i and j refer to Query and Key positions, and M_{pad} indicates the validity of each timestep, where $m_j \in \{0, 1\}$ signifies whether the jth timestep in Key is valid

$$Attention\,(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (2)$$

$$MHA = Concat\,(head_1, head_2, \ldots, head_h)\,W^O \quad (3)$$

$$Masked\ Attention\,(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}} + M_{pad}\right)V, \left(M_{pad}\right)_{ij} = \begin{cases} 0, & m_j = 0 \\ -\infty, & m_j = 1. \end{cases} \quad (4)$$

TABLE II
MAPPING RELATIONSHIP BETWEEN EXISTING PRODUCTS AND SIX LAND COVER TYPES IN THIS STUDY

Dataset	From	To
China Land Cover	Water, Wetland	Water
	Forest, Shrub	Woodland
	Grassland	Grassland
	Cropland	Cropland
	Impervious	Impervious surface
	Barren	Bare
	Snow/Ice	-
ESA WorldCover	Permanent water bodies, Herbaceous wetland	Water
	Tree cover, Shrubland	Woodland
	Grassland	Grassland
	Cropland	Cropland
	Built-up	Impervious surface
	Bare/sparse vegetation	Bare
	Snow and ice, Mangroves, Moss and lichen	-
Dynamic World	water, flooded_vegetation	Water
	trees, shrub_and_scrub	Woodland
	grass	Grassland
	crops	Cropland
	built	Impervious surface
	bare	Bare
	snow_and_ice	-
Esri Land Cover	Water, Flooded Vegetation	Water
	Trees	Woodland
	Rangeland	Grassland
	Crops	Cropland
	Built Area	Impervious surface
	Bare Ground	Bare
	Snow/Ice, Clouds	-

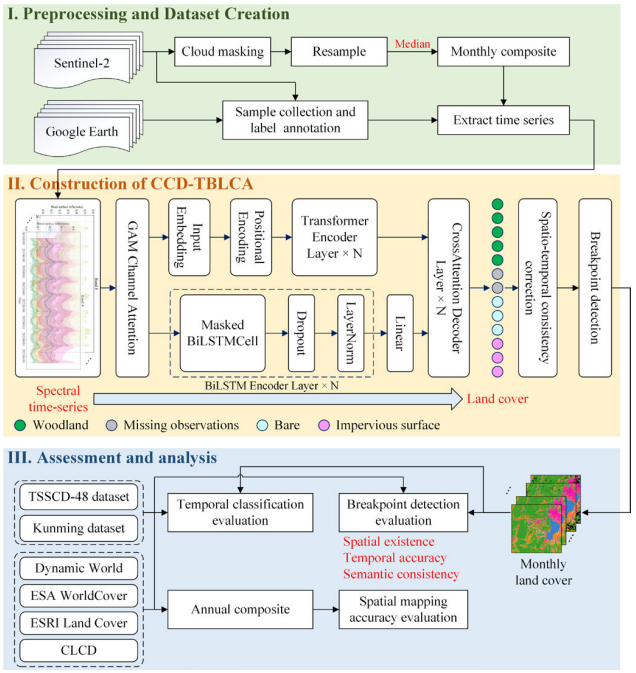


Fig. 4. Technical approach of this study and the model structure of CCD-TBLCA.

BiLSTM formed the second encoder branch and extracted local dynamics to refine breakpoint localization. In (5), LSTM updates the memory cell C_t and hidden state H_t through

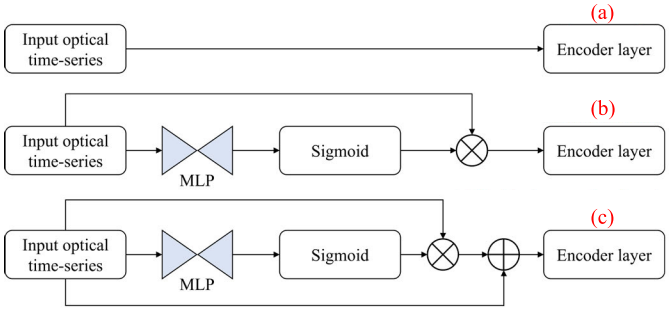
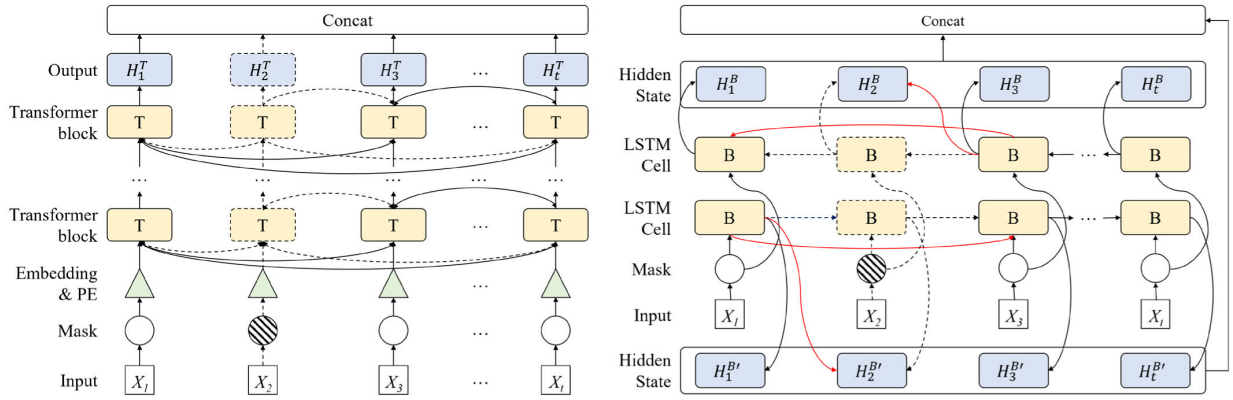


Fig. 5. Principles of direct input, original GAM channel attention, and residual connection-based channel attention. (a) Direct input. (b) Original GAM channel attention. (c) Residual connection-based channel attention.

the forget gate F_t, input gate I_t, and output gate O_t. The candidate memory cell $\tilde{C}_t$ is determined by the current input X_t and the previous hidden state H_{t-1}. W_{xc}, W_{hc}, and b_c are learnable parameters. C_t combines C_{t-1} and $\tilde{C}_t$, where $\odot$ denotes element-wise product, and H_t represents the historical information

$$\begin{cases} \tilde{C}_t = \tanh\ (X_t W_{xc} + H_{t-1} W_{hc} + b_c) \\ C_t = F_t \odot C_{t-1} + I_t \odot \tilde{C}_t \\ H_t = O_t \odot \tanh\ (C_t). \end{cases} \tag{5}$$

Missing observations will break the RNNs' learned sequential dependencies [50]. In Fig. 6(b), the BiLSTM encoder skips



(a) Masked Transformer Encoder (b) Masked BiLSTM Encoder

Fig. 6. Mechanism of the encoder layer for handling missing values. Dashed lines indicate timesteps excluded from gradient propagation. (a) Masked transformer encoder. (b) Masked BiLSTM encoder.

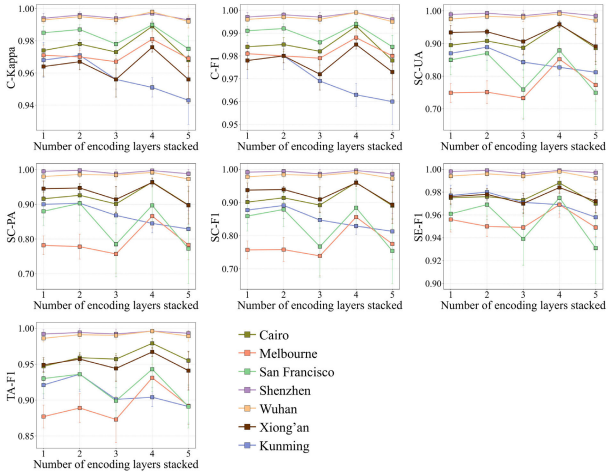


Fig. 7. Parameter sensitivity analysis results for the number of stacked layers N in the transformer and BiLSTM encoding layers across all cities.

missing observations under the control of a mask layer. If an LSTM cell receives a padding value, its hidden state is replaced by the nearest valid value in the history. Masked timesteps do not contribute to the loss. The encoding information $F_B \in \mathbb{R}^{L\times 2d}$ is obtained by concatenating $H_t^B \in \mathbb{R}^d$ and $H_t^{B'} \in \mathbb{R}^d$. Such a mechanism ensures that memory transfer remains uncontaminated by padding noise.

After stacking transformer and BiLSTM encoder blocks $N\times$, F_T and F_B are fed into a multihead CA decoder. A linear projection reduces the dimension of F_B from $\mathbb{R}^{L\times 2d}$ to $\mathbb{R}^{L\times d}$ to match F_T. The number of stacking layers N can be flexibly adjusted based on specific regions and computational costs. The results of the parameter sensitivity analysis for N across all cities are shown in Fig. 7. To ensure consistency in the experimental setup, we set N to 4 by default based on the performance of all cities across different evaluation metrics.

3) Decoder: The transformer branch models long-term dependencies, capturing nonlocal patterns across seasons and years. The BiLSTM branch tracks local dynamics and is sensitive to subtle changes and abrupt edges [51]. Their integration can reduce local classification errors and sharpen decision boundaries. To fully combine their strengths, we designed a dual-branch decoder that employed symmetric CA to achieve information fusion in Fig. 8. Each block contains intrabranch self-attention, cross-branch attention, and a feedforward network (FFN). For the BiLSTM branch, multihead self-attention is first applied by (6). Subsequently, the BiLSTM features serve as Query, while the transformer encoding information acts as Key and Value in (7). Finally, FFN extracts deeper features by (8) and (9). The transformer branch follows the same procedure. After stacking N blocks, we concatenate the outputs and apply a fully connected layer to predict per-timestep categories

$$F_B^{SA(k)} = LN\left(F_B^{(k-1)} + MHA_B^{self}\left(F_B^{(k-1)}, F_B^{(k-1)}, F_B^{(k-1)}\right)\right) \tag{6}$$

$$F_B^{CA(k)} = LN\left(F_B^{SA(k)} + MHA_{B\leftarrow T}^{cross}\left(F_B^{SA(k)}, F_T, F_T\right)\right) \tag{7}$$

$$FFN_B^{(k)}(x) = \sigma\left(xW_{B,1}^{(k)} + b_{B,1}^{(k)}\right)W_{B,2}^{(k)} + b_{B,2}^{(k)} \tag{8}$$

$$F_B^{(k)} = LN\left(F_B^{CA(k)} + FFN_B^{(k)}\left(F_B^{CA(k)}\right)\right) \tag{9}$$

where $F_\cdot^{SA(k)}$ and $F_\cdot^{CA(k)}$ denote the outputs of self-attention and CA at the kth block, respectively, $F_\cdot^{(k)}$ denotes the outputs of the kth decoder block, $LN(\cdot)$ and $\sigma(\cdot)$ denote the layer normalization and ReLU activation function, $W_\cdot^{(k)}$ and $b_\cdot^{(k)}$ are learnable parameters.

4) Postprocessing and Breakpoint Definition: To reduce salt-and-pepper noise and label oscillations, we applied a spatiotemporal consistency correction as postprocessing, a widely used approach in land cover mapping [52], [53]. A 3×3 2-D window and a 1-D window of length five reassigned the center to the majority category.

For each pixel, a breakpoint was defined as the later timestep when a land cover change occurred. If the category changed between timestep t_i and timestep t_{i+1}, the breakpoint was t_{i+1}. Based on this, we derived several sequences containing breakpoint timesteps. The first breakpoints formed the breakpoint map reflecting initial land cover changes. The maximum length of the breakpoint sequences equaled the number of land cover changes.

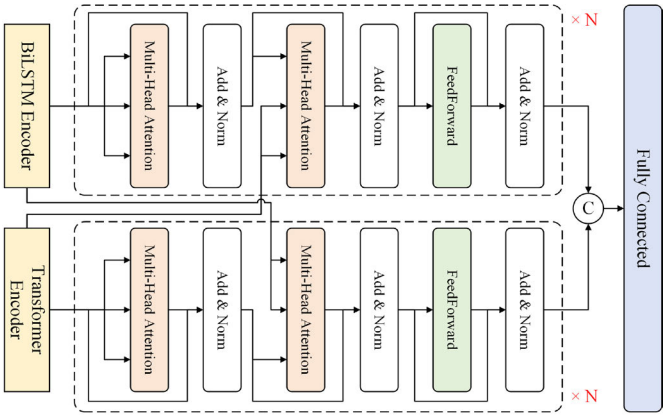


Fig. 8. Structure of a multihead CA decoder fusing dual-branch encoding information.

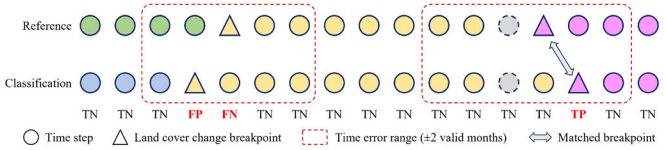


Fig. 9. Evaluation criterion for breakpoint detection based on semantic consistency. Different colors represent distinct land cover types.

B. Evaluation Metrics

CCD-TBLCA produced per-timestep land cover labels and aimed to determine when and what type of change occurred. Therefore, we employed two temporal classification metrics and five breakpoint detection metrics. Using 1-D time series as the computational unit, we calculated Cohen's kappa (C-Kappa) and F1-score (C-F1) to evaluate temporal classification performance by the following equation:

$$F1^C = \frac{1}{N}\sum_{i}^{N}\left(\frac{1}{K}\sum_{k}^{K}\frac{2TP_k^{(i)}}{2TP_k^{(i)}+FP_k^{(i)}+FN_k^{(i)}}\right) \tag{10}$$

$$p_o^{(i)} = \frac{1}{L^{(i)}}\sum_{k}^{K} n_{kk}^{(i)} \tag{11}$$

$$p_e^{(i)} = \frac{1}{\left(L^{(i)}\right)^2}\sum_{k}^{K}\left(\left(\sum_{m} n_{km}^{(i)}\right)\left(\sum_{j} n_{jk}^{(i)}\right)\right) \tag{12}$$

$$Kappa^C = \frac{1}{N}\sum_{i}^{N}\frac{p_o^{(i)}-p_e^{(i)}}{1-p_e^{(i)}} \tag{13}$$

where $N, L^{(i)}$, and K represent the number of samples, the number of valid months in sequence i, and the number of categories, respectively; $\mathrm{TP}_k^{(i)}$, $\mathrm{FP}_k^{(i)}$, and $\mathrm{FN}_k^{(i)}$ denote the true positives, false positives, and true negatives for category k in sequence, i, m, and j represent specific land cover type, and $n_{\cdot,\cdot}^{(i)}$ denotes the values for each element in the confusion matrix.

Previous studies evaluated breakpoint detection from spatial existence and temporal accuracy. However, counting breakpoints and checking their timing cannot represent real geographic scenarios if land cover semantics are ignored. Therefore, we proposed a stricter metric based on semantic consistency, as illustrated in Fig. 9. For the reference data incorporating land cover changes, a predicted breakpoint is treated as TP only when its time difference from the truth falls within ±2 valid months and the categories before and after the change match. Under this rule, we introduced user's accuracy (UA), producer's accuracy (PA), and F1-score in (14)–(16), denoted as SC-UA, SC-PA, and SC-F1, respectively. We also used F1-score to evaluate breakpoint's spatial existence (SE-F1) and temporal accuracy (TA-F1)

$$UA^{SC} = \frac{1}{N'}\sum_{i}^{N'}\frac{TP^{(i)}}{TP^{(i)}+FP^{(i)}} \tag{14}$$

$$PA^{SC} = \frac{1}{N'}\sum_{i}^{N'}\frac{TP^{(i)}}{TP^{(i)}+FN^{(i)}} \tag{15}$$

$$F1^{SC} = \frac{1}{N'}\sum_{i}^{N'}\frac{2TP^{(i)}}{2TP^{(i)}+FP^{(i)}+FN^{(i)}} \tag{16}$$

where N' represents the number of changing samples. $\mathrm{TP}^{(i)}, \mathrm{FP}^{(i)}$, and $\mathrm{FN}^{(i)}$ denote the TP, FP, and FN in sequence i based on breakpoints, respectively. TA-F1 was computed over all time series to assess temporal alignment, whereas SC-F1 was confined to reference data with changes to evaluate whether CCD-TBLCA could capture changes consistent with the real-world. To reduce randomness, we shuffled the datasets into training, validation, and test sets, independently trained CCD-TBLCA ten times, and computed the mean and standard deviation of each metric.

TABLE III
HYPERPARAMETER SEARCH RANGE FOR RFC

Hyperparameters	Values
n_estimators	{100, 150, 200, 250}
max_depth	{5, 10, 15, 20}
max_features	{sqrt, log2}
min_samples_split	{2, 5, 10}
criterion	{gini, entropy}

C. Benchmark

To demonstrate the effectiveness of CCD-TBLCA, we compared several common methods. One category uses time series as inputs, including CCDC [10] and TSSCD [30]. CCDC was run on GEE with default hyperparameters and the same detection bands as CCD-TBLCA. TSSCD with a convolution channel configuration of [128, 256, 512, 1024] required seamless sequences. Padding values were linearly interpolated, and labels were modified by nearest-neighbor. Another category uses all bands at each timestep, including multilayer perception (MLP) and random forest classifier (RFC). We set the number of neurons in MLP to [64, 128, 256] and optimized RFC by GridSearchCV based on Table III. 2-D semantic segmentation networks were not considered due to the dataset format.

D. Experimental Configuration

All models were run on Windows 11 with Python 3.12.9 and PyTorch 2.5.1. Preprocessing and postprocessing were

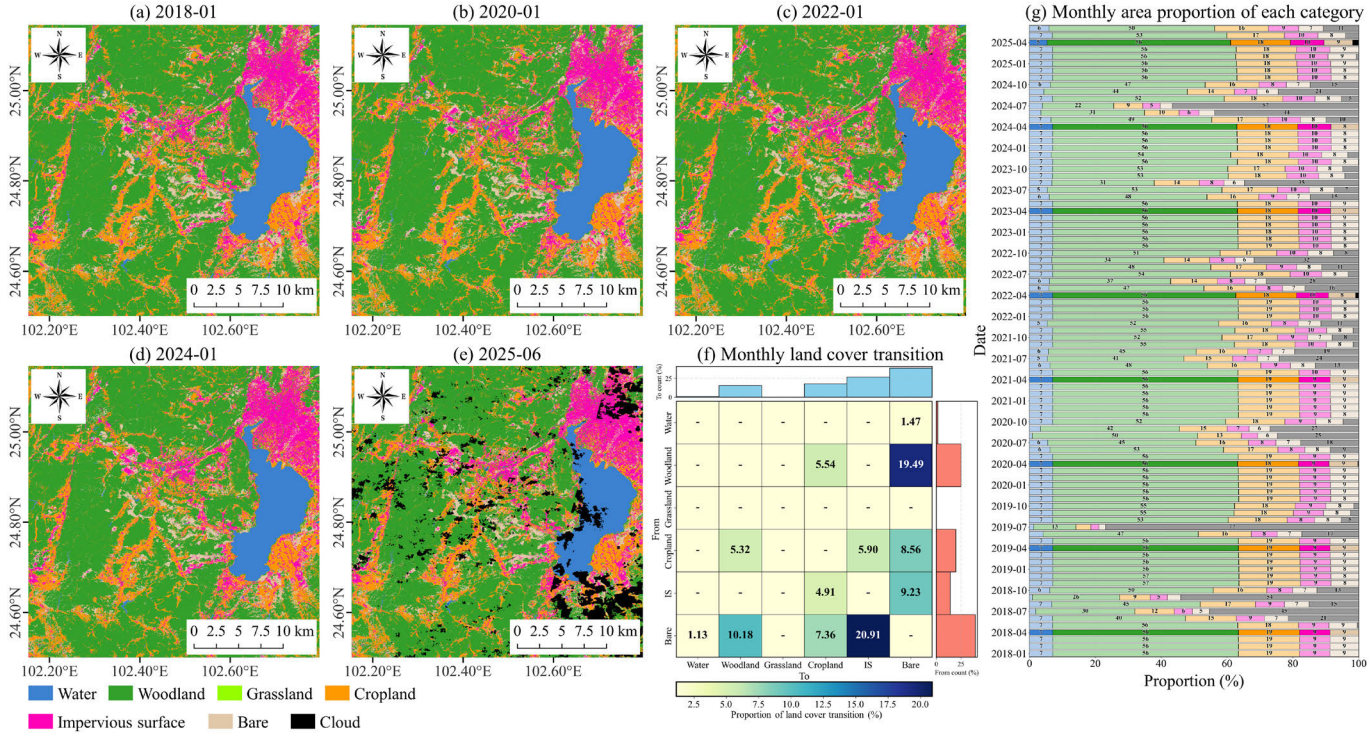


Fig. 10. (a)–(e) Monthly land cover classification results within the Kunming study area. (f) Land cover transition matrix using all pixels. (g) Changes in the area proportion of each category.

performed using an Intel i7-12700K CPU, while model training was accelerated using an NVIDIA GeForce RTX 3090 GPU. In CCD-TBLCA, the transformer embedding and BiLSTM hidden state had 128 dimensions, with eight heads in MHA. The encoder and decoder blocks were each stacked four times. The batch size was 64. Model optimization employed the Adaptive Moment Estimation optimizer with a learning rate of 0.001 and a cross-entropy loss function in (17). An early stopping mechanism with patience set to ten was implemented based on C-F1.

$$Loss_{CE} = \frac{1}{\left(\sum_i \sum_{l=1}^{L} \left(1 - M_{i,l}\right)\right)} \sum_i \sum_{l=1}^{L} \left(1 - M_{i,l}\right) \times \left(-\sum_c y_{i,l,c} \log p_{i,l,c}\right) \tag{17}$$

where L and C represent the time-series length and the number of categories, respectively, while $M_{i,l} \in \{0, 1\}$ indicates whether the sample i is valid at the lth timestep, with a value of one meaning the current timestep is invalid. At the lth timestep, if the true category of the sample i is equal to c, $y_{i,l,c}$ takes one, otherwise it takes zero, and $p_{i,l,c}$ denotes the probability that the sample i is classified as c.

IV. Results

A. Monthly Land Cover Mapping

After obtaining monthly land cover maps, we focused not only on spatial mapping but also on land cover changes. In Fig. 10(a)–(e), the Kunming study area is primarily composed of woodland and cropland, with impervious surface mainly distributed northeast of Dian Lake. The proportion of transitions between different categories is shown in Fig. 10(f). Considering classification errors, we excluded changes with proportions less than 1%. The dominant transitions are woodland to bare land and bare to impervious surface, each accounting for approximately 20%, associated with mining and urban development. Additionally, cropland's gain and loss, conversion of bare land to woodland, and transformation of impervious surface into bare land also account for a significant proportion. Monthly results show stable area proportions across all categories in Fig. 10(g). Excluding months with heavy cloud cover, the proportion of various land cover types fluctuates by only 1%.

B. Accuracy Assessment

1) Breakpoint Detection Principle: For breakpoint detection, CCD-TBLCA and TSSCD adopted change detection after classification, whereas CCDC first detected breakpoints before classification. Fig. 11 qualitatively compares these principles using two sites in Kunming: woodland transforming to bare land and then to impervious surface, and water converting to bare land.

In Fig. 11(a), the sample undergoes two distinct change processes, typical of forest degradation and urban expansion. CCDC detected only the change event from woodland to bare land, but missed the rapid construction-related change. TSSCD and CCD-TBLCA captured these two critical breakpoints. Compared to TSSCD, CCD-TBLCA exhibited greater differences in probability distributions, reflecting more obvious land cover transitions. This indicates a more agile and stable response to abrupt events.

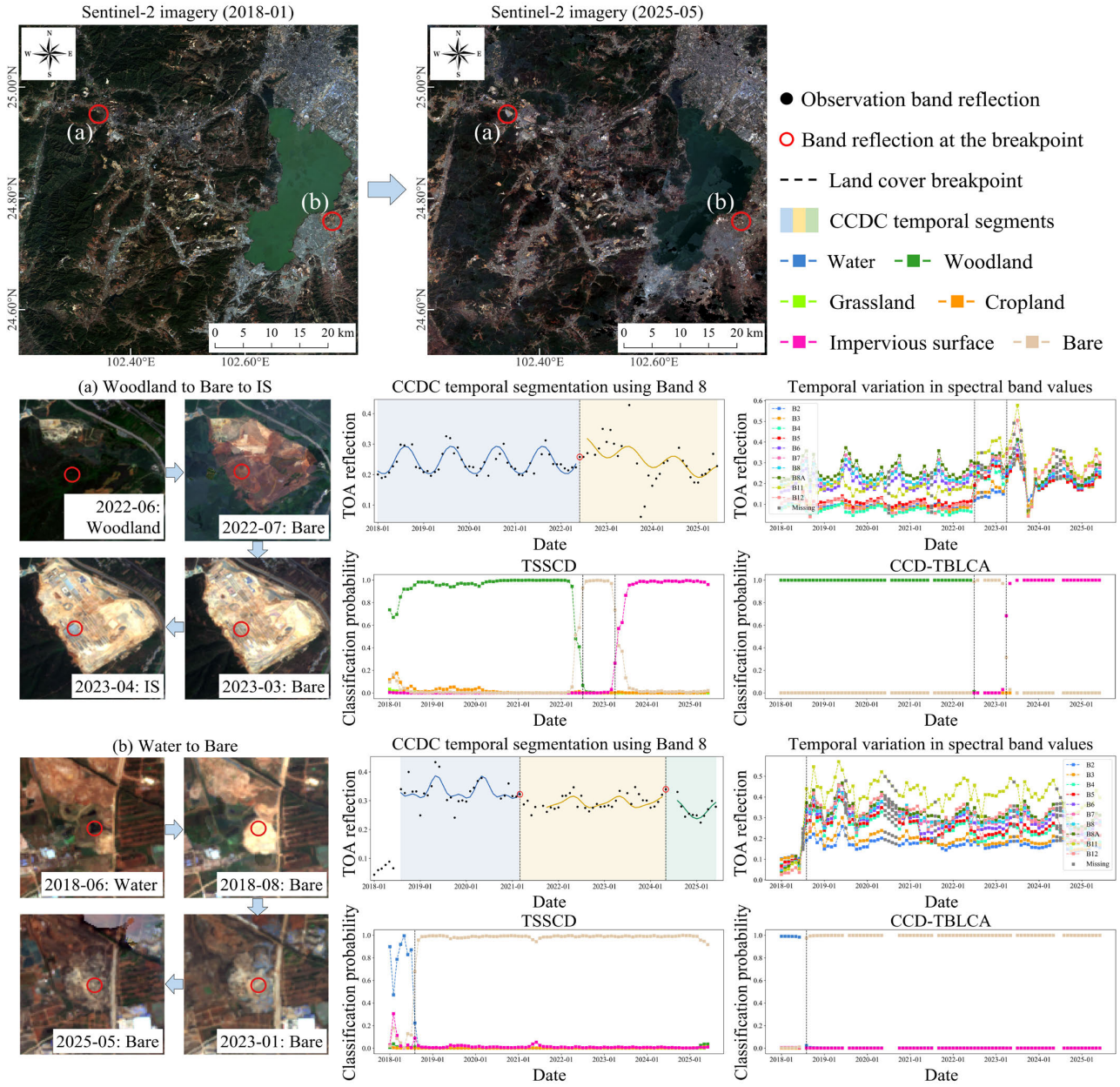


Fig. 11. Model principles of CCDC, TSSCD, and CCD-TBLCA in breakpoint detection. To better display the time-series curves of spectral values across different bands, we employed linear interpolation to fill in missing observations when plotting temporal variation in each band, labeling these as "Missing." It is important to note that CCD-TBLCA still used the time-series data containing these missing observations as input. (a) Woodland to bare land to IS. (b) Water to bare land.

In Fig. 11(b), the spectral curve shows a clear transition from water to bare land. CCDC failed to detect the early change, instead identifying two fake breakpoints. During initialization, CCDC performed spectral stability testing, making early changes highly susceptible to omission. After the sample transitioned to bare land, alterations in surface color and brightness might cause CCDC to identify fake breakpoints. In contrast, CCD-TBLCA captured land cover change with greater accuracy and reliability.

2) Spatial Mapping Details: To access the performance of different continuous change detection methods in spatial mapping, we generated monthly results for Kunming and conducted a detailed comparison around several typical areas in Fig. 12. In the greenhouse-dominated cropland area southeast of Dian Lake [Fig. 12(a)], RFC, MLP, CCDC, and TSSCD misclassified large portions of cropland as impervious surface. In contrast, CCD-TBLCA better separated these categories and extracted precise road structures.

Fig. 12(b) depicts a bare region resulting from mining. The breakpoint map clearly illustrates the conversion to bare land. RFC and MLP performed the worst, alongside CCDC, misclassifying significant amounts of woodland and bare land as cropland, blurring the boundaries between natural vegetation and agricultural land. Furthermore, both RFC and MLP, along with TSSCD, confused large areas of bare land with impervious surfaces. CCD-TBLCA best preserved the segmentation boundaries among woodland, cropland, impervious surface, and bare land. In the mountainous region [Fig. 12(c)],

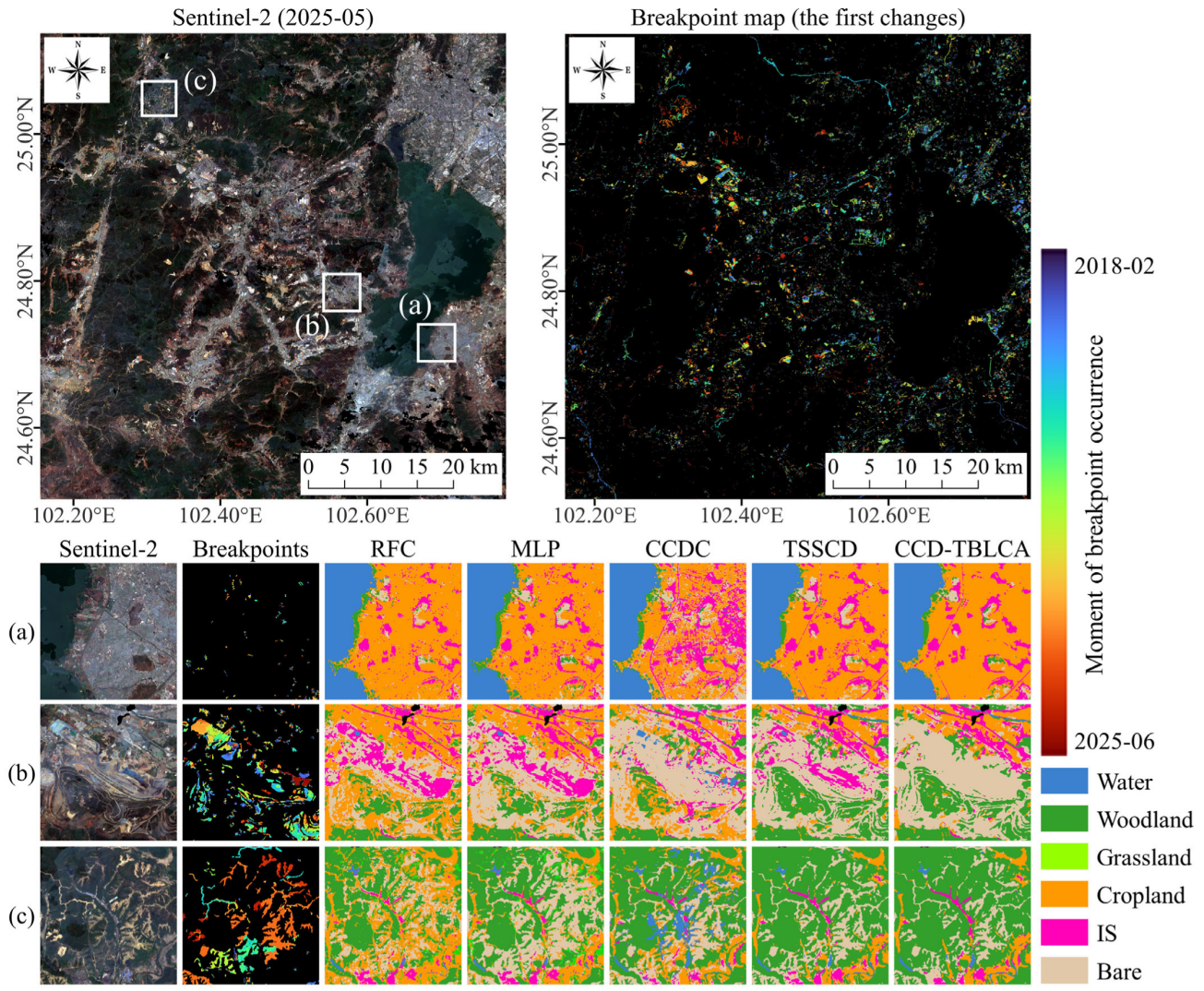


Fig. 12. Detailed comparison of different methods for spatial mapping in Kunming, with a breakpoint map generated by CCD-TBLCA. In the upper section, the left panel displays a true-color Sentinel-2 L1C monthly composite image, while the right panel shows a map derived from the first batch of land cover change breakpoints. The gradient-colored patches indicate the moment of land cover change occurrence. (a) Denotes the greenhouse-dominated cropland area. (b) Denotes the bare region resulting from mining. (c) Denotes the deforestation activities.

CCD-TBLCA and TSSCD accurately detected deforestation, while RFC mislabeled extensive bare land as cropland, MLP exhibited scattered classification errors, and CCDC incorrectly classified woodland as water.

3) Quantitative Assessment: We then quantitatively compared CCD-TBLCA with four benchmarks across different metrics, as summarized in Fig. 13. The bar height and error bars represent the mean and standard deviation, respectively. Detailed results for Cairo and Kunming are shown in Table IV. For temporal classification, CCD-TBLCA achieved the best C-Kappa and C-F1 across all cities, surpassing CCDC, MLP, and RFC, and improving over TSSCD by 3.34% and 2.16% in Cairo and by 6.37% and 4.22% in Kunming. CCD-TBLCA also exhibited smaller variations, indicating greater stability.

For breakpoint detection, CCD-TBLCA outperformed in both spatial existence (SE-F1) and temporal accuracy (TA-F1) across all cities. In Kunming, its TA-F1 exceeded all benchmarks by at least 21.83%, highlighting its advantage in localizing changes. Regarding semantic consistency, CCD-TBLCA achieved the best accuracy across all cities, with SC-F1 typically at least 15% higher than others. TSSCD's SC-F1 significantly outperformed RFC, MLP, and CCDC, contrasting with its weaker performance in TA-F1. This highlights the necessity of incorporating semantic consistency when assessing breakpoint detection.

C. Comparison With Existing Products

Numerous large-scale land cover products, including the annually updated CLCD, Esri Land Cover, ESA WorldCover, and the near-real-time Dynamic World, have been widely applied in land cover analysis [54], [55]. Using test sets from each city, we compared their spatial mapping accuracy with CCD-TBLCA. To harmonize temporal resolution, we applied a mode composite to Dynamic World and our monthly-updated results to generate annual maps, and evaluated all products

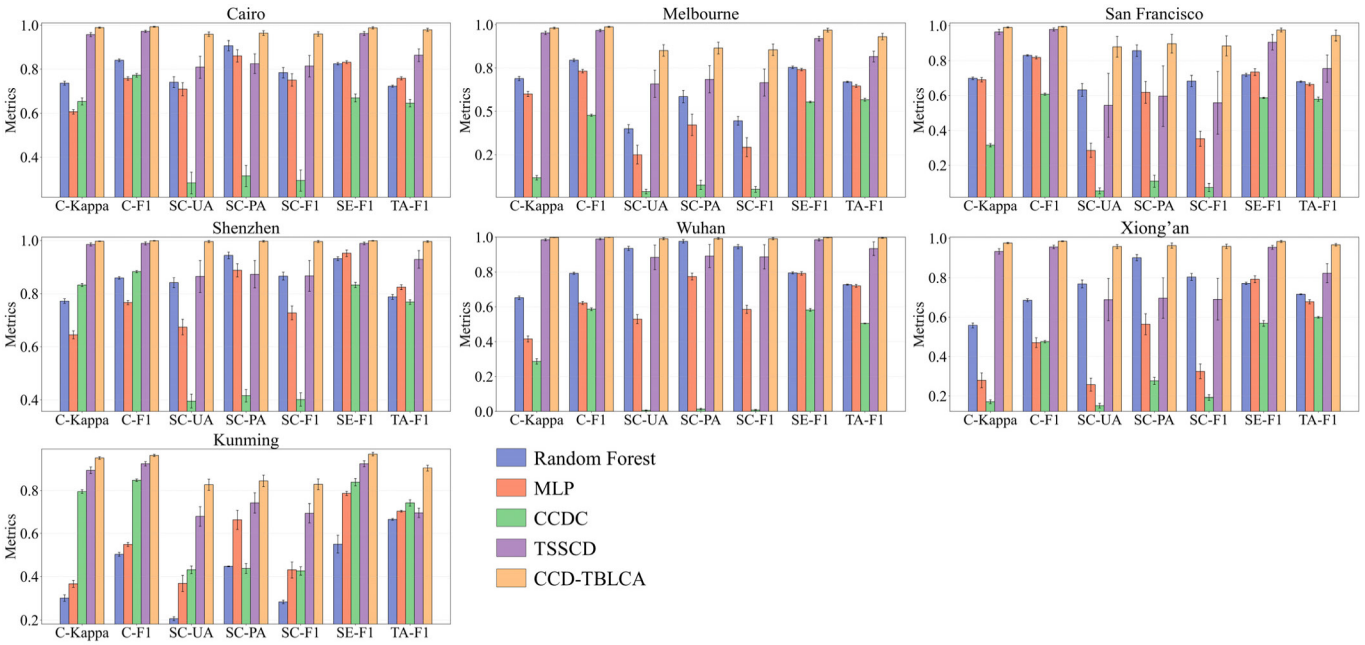


Fig. 13. Accuracy comparison of different methods for temporal classification and breakpoint detection across all cities.

TABLE IV
DETAILED COMPARISON OF EVALUATION METRICS FOR DIFFERENT METHODS IN CAIRO AND KUNMING

Area	Metrics	Random Forest	MLP	CCDC	TSSCD	CCD-TBLCA
Cairo	C-Kappa	0.737±0.009	0.606±0.010	0.653±0.016	0.957±0.009	**0.989±0.003**
	C-F1	0.841±0.006	0.758±0.008	0.772±0.008	0.972±0.005	**0.993±0.002**
	SC-UA	0.741±0.025	0.709±0.030	0.282±0.049	0.809±0.050	**0.959±0.010**
	SC-PA	0.907±0.023	0.861±0.028	0.314±0.048	0.825±0.045	**0.964±0.011**
	SC-F1	0.784±0.023	0.751±0.028	0.293±0.048	0.814±0.049	**0.960±0.010**
	SE-F1	0.825±0.006	0.832±0.007	0.669±0.018	0.962±0.009	**0.988±0.005**
	TA-F1	0.723±0.005	0.759±0.007	0.645±0.017	0.864±0.029	**0.979±0.007**
Kunming	C-Kappa	0.300±0.015	0.366±0.016	0.795±0.009	0.894±0.015	**0.951±0.006**
	C-F1	0.503±0.009	0.549±0.009	0.848±0.006	0.924±0.010	**0.963±0.005**
	SC-UA	0.206±0.009	0.368±0.037	0.430±0.018	0.679±0.046	**0.827±0.026**
	SC-PA	0.446±0.002	0.663±0.045	0.436±0.023	0.742±0.047	**0.845±0.027**
	SC-F1	0.282±0.008	0.430±0.037	0.425±0.019	0.694±0.045	**0.829±0.025**
	SE-F1	0.550±0.041	0.786±0.010	0.839±0.017	0.924±0.014	**0.969±0.008**
	TA-F1	0.665±0.004	0.704±0.004	0.742±0.014	0.696±0.022	**0.904±0.013**

TABLE V
MODULE COMPOSITION OF DIFFERENT MODEL VERSIONS IN THE ABLATION EXPERIMENT

Model version	BiLSTM encoder	Transformer encoder	Cross-Attention decoder	Original GAM channel attention	Residual connection-based channel attention
BiLSTM	√	×	×	×	√
Transformer	×	√	×	×	√
CCD-TBL	√	√	×	×	√
GAM-None	√	√	√	×	×
GAM-Original	√	√	√	√	×
CCD-TBLCA	√	√	√	×	√

TABLE VI
DETAILED COMPARISON OF EVALUATION METRICS FOR DIFFERENT MODEL VERSIONS IN CAIRO AND KUNMING

Area	Metrics	BiLSTM	Transformer	CCD-TBL	GAM-None	GAM-Original	CCD-TBLCA
Cairo	C-Kappa	0.984±0.005	0.981±0.004	0.980±0.006	0.974±0.009	0.977±0.005	**0.989±0.003**
	C-F1	0.990±0.003	0.988±0.003	0.988±0.004	0.983±0.006	0.985±0.003	**0.993±0.002**
	SC-UA	0.901±0.036	0.858±0.028	0.896±0.034	0.900±0.034	0.914±0.025	**0.959±0.010**
	SC-PA	0.912±0.036	0.878±0.025	0.909±0.030	0.913±0.036	0.926±0.019	**0.964±0.011**
	SC-F1	0.904±0.036	0.863±0.027	0.900±0.033	0.904±0.034	0.917±0.023	**0.960±0.010**
	SE-F1	0.985±0.005	0.983±0.004	0.981±0.008	0.973±0.008	0.978±0.007	**0.988±0.005**
	TA-F1	0.951±0.017	0.921±0.016	0.938±0.018	0.957±0.013	0.961±0.016	**0.979±0.007**
Kunming	C-Kappa	0.949±0.004	0.947±0.007	0.946±0.004	0.943±0.006	0.937±0.008	**0.951±0.006**
	C-F1	0.962±0.003	0.961±0.006	0.960±0.004	0.957±0.004	0.954±0.006	**0.963±0.005**
	SC-UA	0.804±0.033	0.802±0.031	0.809±0.031	0.806±0.032	0.783±0.027	**0.827±0.026**
	SC-PA	0.819±0.032	0.827±0.029	0.830±0.028	0.827±0.034	0.810±0.025	**0.845±0.027**
	SC-F1	0.806±0.032	0.807±0.030	0.810±0.030	0.809±0.032	0.789±0.027	**0.829±0.025**
	SE-F1	0.964±0.005	0.965±0.008	0.960±0.007	0.966±0.010	0.959±0.011	**0.969±0.008**
	TA-F1	0.890±0.010	0.879±0.017	0.890±0.014	0.894±0.019	0.878±0.016	**0.904±0.013**

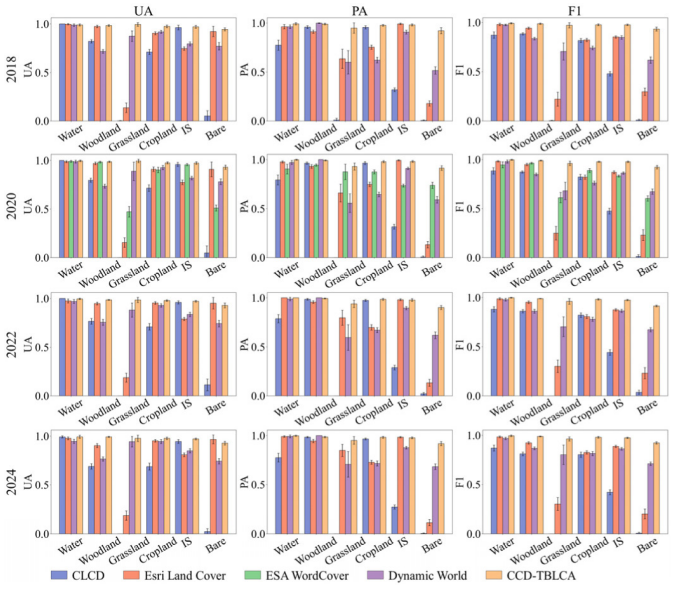


Fig. 14. Spatial mapping accuracy comparison between CCD-TBLCA and existing products in Kunming in 2018, 2020, 2022, and 2024.

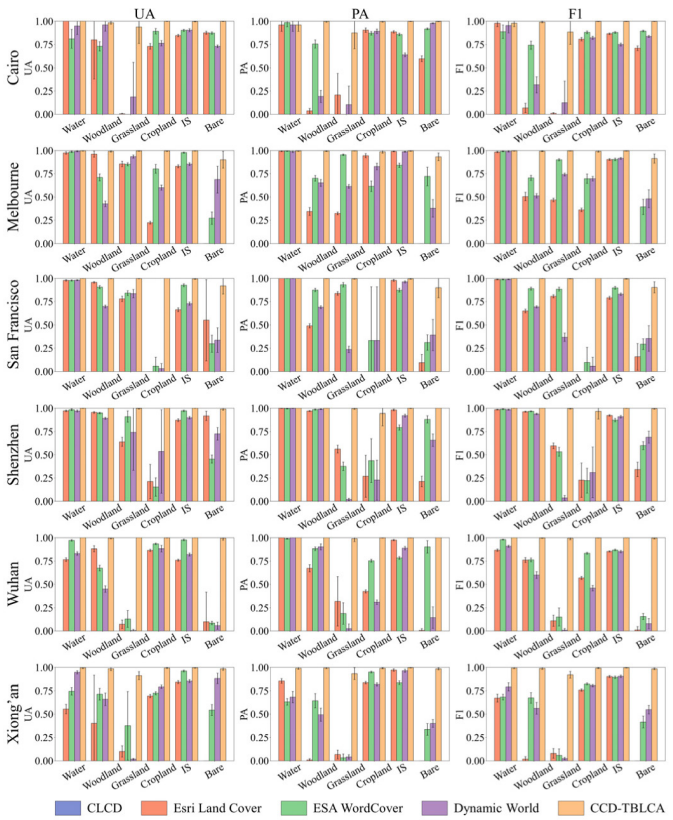


Fig. 15. Spatial mapping accuracy comparison between CCD-TBLCA and existing data products in more cities in 2020.

using UA, PA, and F1-score. Taking Kunming as an example, CCD-TBLCA achieved the highest F1-score (generally above 0.9) and the smallest variability across most categories in Fig. 14. Others exhibited significant shortcomings in specific categories, with accuracy lower than CCD-TBLCA. Validation results for other cities in 2020 are detailed in Fig. 15, which similarly confirm CCD-TBLCA's superior accuracy.

It is worth noting that in some cities, existing data products have unusually poor metrics, such as UA for several land cover types, while the CCD-TBLCA classification results are better. In addition to the misclassification of the existing products themselves, there may be two other reasons. First, differences in category definitions across existing data products might create errors in the process of category remapping based on Table II. For example, we set bare/sparse vegetation in ESA WordCover to bare land, which would lead to grassland with sparse vegetation cover being confused with real bare

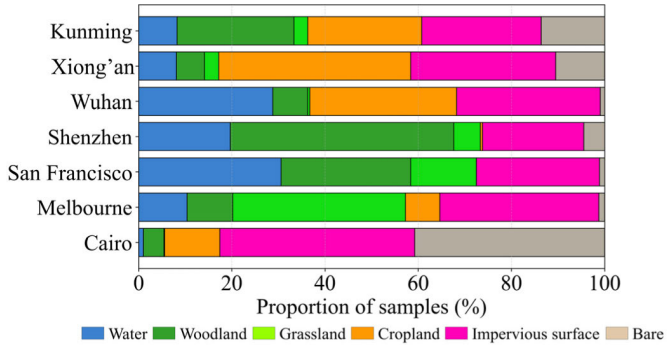


Fig. 16. Proportion of samples of different land cover types within all cities used to assess spatial mapping accuracy in 2020. Annual labels were obtained by taking the plurality of land cover types for all months in 2020.

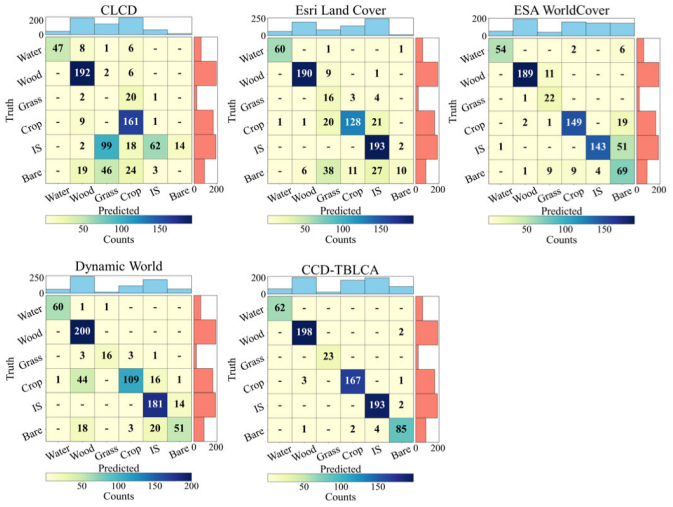


Fig. 17. Confusion matrices for land cover classification in 2020, comparing CCD-TBLCA with existing products. Wood, Grass, Crop, and IS represent woodland, grassland, cropland, and impervious surface, respectively.

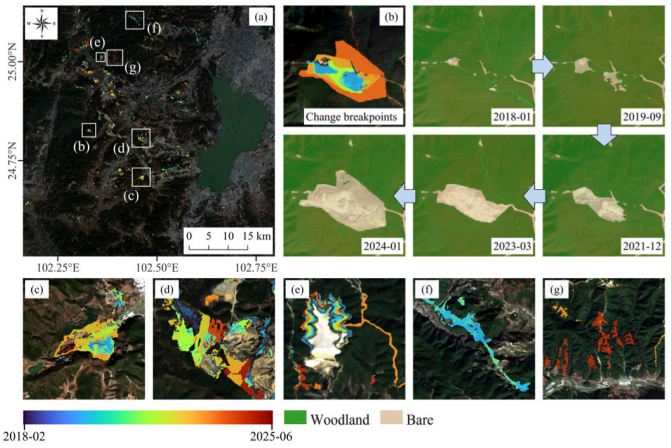


Fig. 18. Detailed characterization of the spatiotemporal dynamics of woodland. (a) Shows the first breakpoint map. (b)–(e) Represent open-air mining. (f) Represents road construction. (g) Represents deforestation.

land, resulting in underestimated UA values for bare land. Second, and perhaps more critically, the six land cover types were highly imbalanced across cities, with some categories occupying only a very small area. Categories with tiny proportions of area often had only a few validation samples. Combined with inherent classification errors in the existing products, this could easily lead to unstable UA values as well as the appearance of abnormally low values. For example, ESA WordCover 2020 shows a low proportion of grassland and bare land in Wuhan (Fig. 1), and accordingly, Fig. 16 shows a small proportion of validation samples for grassland and bare land, both of which have tiny UA values in the existing data products. Cairo, on the other hand, has a large area of bare land but a small area of grassland, resulting in the existing products having a more stable UA for classification of bare land but a poor UA for grassland. Therefore, when validating the classification performance of CCD-TBLCA for grassland, it is most appropriate to compare only the two cities with high grassland coverage: Melbourne and San Francisco. The results in Fig. 15 show that our model achieved better spatial mapping accuracy.

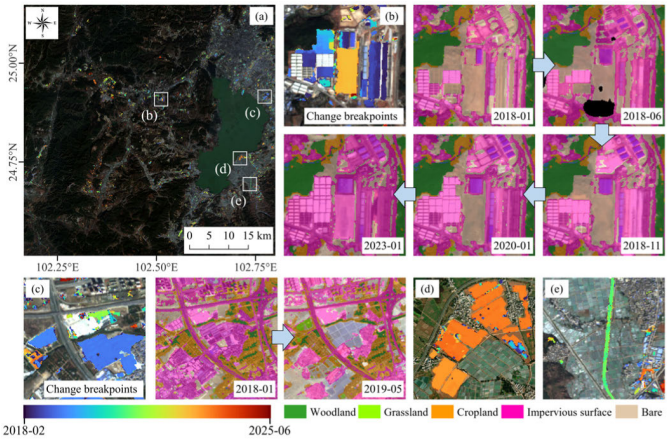


Fig. 19. Detailed characterization of the spatiotemporal dynamics of impervious surface. (a) Shows the first breakpoint map. (b), (d), and (e) Represent new construction. (c) Represents building removal.

To further analyze the discrimination capability of CCD-TBLCA for each category, we generated annual confusion matrices across all products. Taking Kunming as an example, the results in 2020 are shown in Fig. 17. CLCD performed worst, misclassifying numerous impervious surface samples and bare land samples as grassland while showing zero recognition of grassland and bare land. Esri Land Cover and Dynamic World showed noticeable misclassifications in cropland and bare land. ESA WorldCover confused impervious surface with bare land. In contrast, CCD-TBLCA demonstrated the best discrimination capability for each category.

D. Tracking of Specific Land Cover Types

As a model for monitoring continuous land cover changes, CCD-TBLCA has been validated in spatial classification and temporal breakpoint detection. However, in many ecological and social applications, attention is often directed to specific targets such as forests or built-up areas. To fully demonstrate its applicability, we analyzed woodland and impervious surface changes in Kunming, which are related to ecological conservation and urban expansion.

Fig. 18 details forest degradation driven by human activities. Fig. 18(a) maps the first occurrence of transitions between

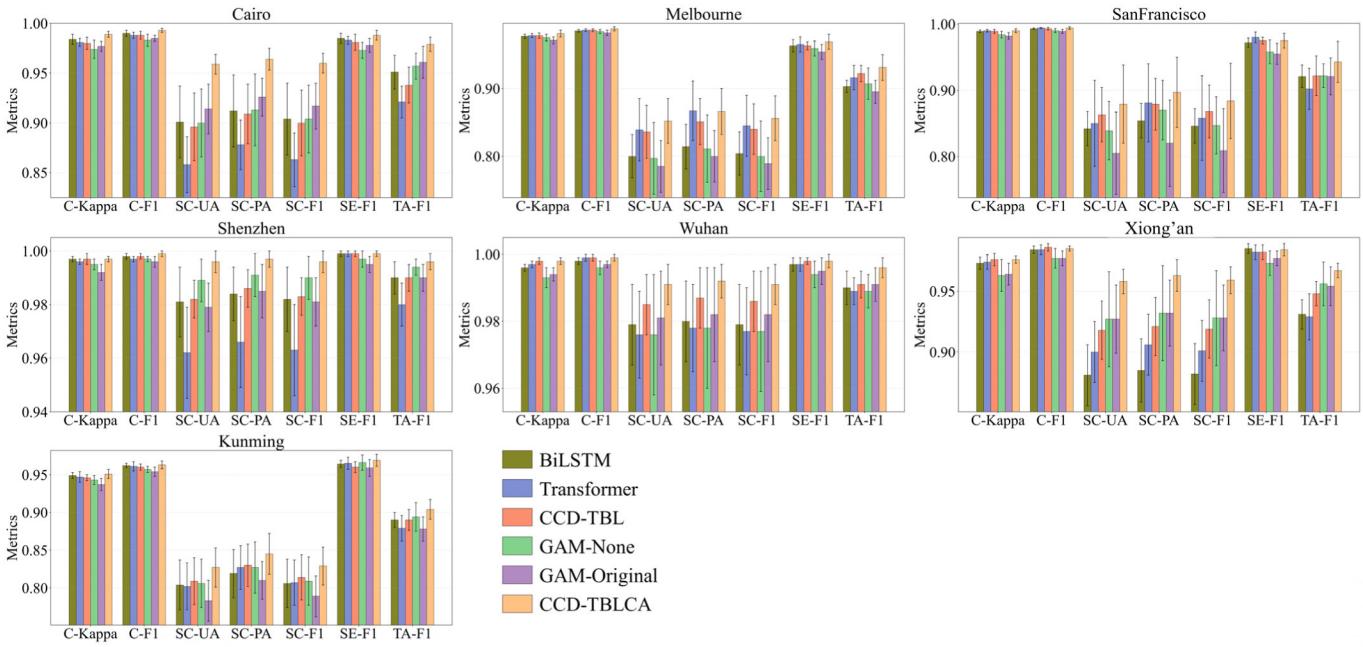


Fig. 20. Accuracy comparison across different model versions across all cities in the ablation experiment.

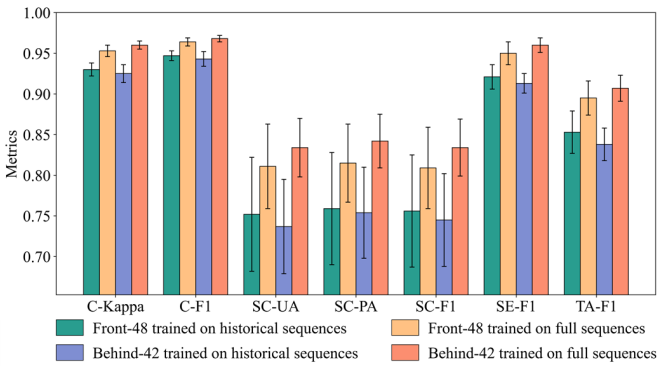


Fig. 21. Accuracy comparison between the 48-month historical sequences and the 42-month latest sequences.

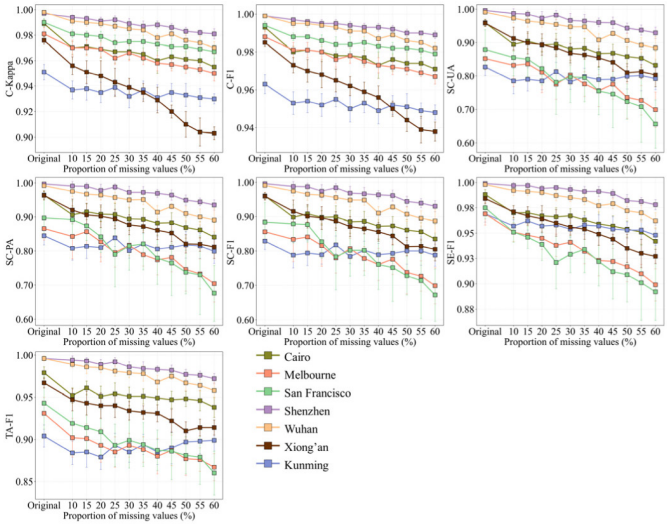


Fig. 22. Impact of missing value proportion in the time series on model accuracy.

woodland and other categories, where the blue to red gradient patches represent breakpoints. Fig. 18(b) shows the expansion of a mining area from 2019 to 2024, during which CCD-TBLCA accurately captured both mine expansion and the resulting forest degradation. Fig. 18(c)–(e) similarly demonstrate its capability to monitor open-air mining. Additionally, forest degradation correlates with new construction and deforestation, as shown in Fig. 18(f) and (g), respectively.

Urban development needs to account for impervious surface as a key category. Fig. 19(a) displays the spatiotemporal distribution of the first impervious surface changes, including new development and removal. Fig. 19(b) traces the continuous construction of a building complex from 2018 to 2023, where CCD-TBLCA accurately captured the staged expansion. Fig. 19(c) illustrates building removal around May 2019. Fig. 19(d) displays the first conversion of cropland to impervious surface, and Fig. 19(e) details new road construction.

CCD-TBLCA can be readily adapted to specialized fields requiring temporal information on specific categories. When focusing on a single category, dataset preparation simplifies to a binary classification problem.

V. Discussion

A. Ablation Experiments

CCD-TBLCA comprises three key modules: a channel attention for inputs, a dual-branch encoder combining transformer and BiLSTM, and a CA decoder for information fusing. To assess the contribution of each component, we designed six versions in Table V. The version lacking the CA decoder employed a simple fully connected classifier.

As shown in Fig. 20, the results indicate that CCD-TBLCA exhibited a smaller accuracy improvement compared to other versions in C-Kappa and C-F1 metrics related to temporal

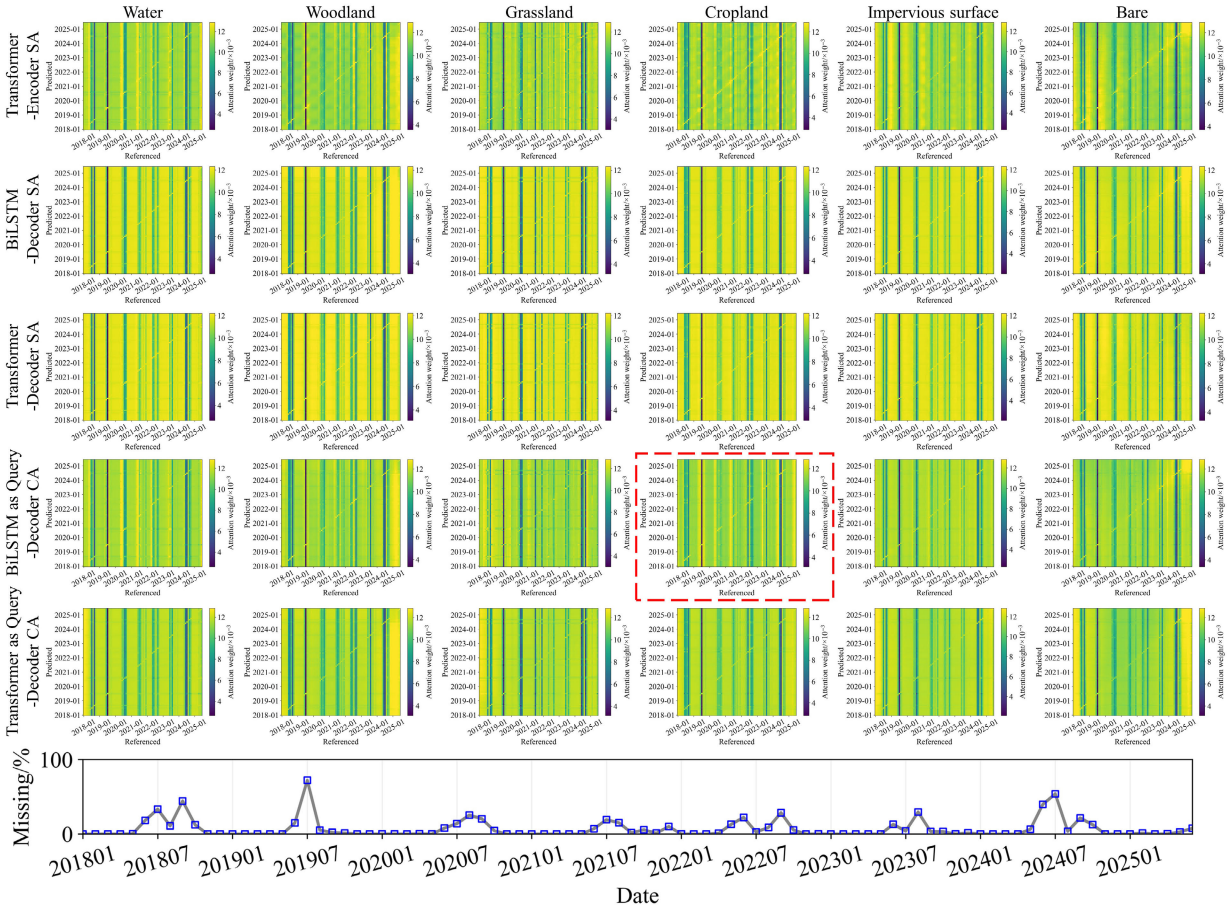


Fig. 23. Each MHA's mean attention weights during the encoding and decoding process (only showing the last layer), along with the proportion of missing observations.

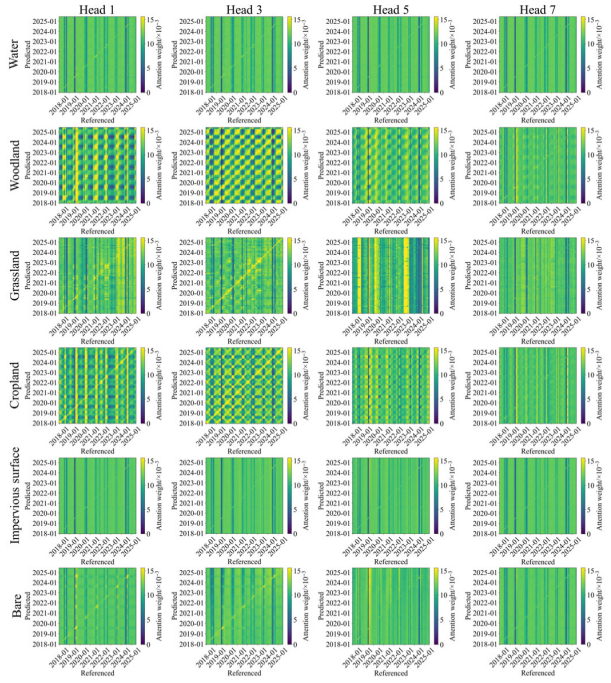


Fig. 24. Attention weights for different heads within MHA in the last transformer encoding layer.

classification, as well as in SE-F1 related to the breakpoint's spatial existence. In most cities, the model incorporating all modules achieved significant improvements in semantic consistency (SC-F1) and breakpoint's temporal accuracy (TA-F1), particularly in Cairo, Shenzhen, Wuhan, and Xiong'an New Area. Detailed comparisons in Cairo and Kunming (Table VI) show that CCD-TBLCA achieved optimal performance across all metrics. Under semantic consistency constraints, CCD-TBLCA achieved SC-F1 improvements ranging from 4.69% to 11.24% in Cairo. In Kunming, SC-F1 improvements were relatively smaller, with a maximum gain of 5.07%. Compared with transformer and BiLSTM, the continuous change detection model that decodes transformer and BiLSTM encoding information through simple concatenation (CCD-TBL) provided intermediate or occasionally superior accuracy, yet remained inferior to CCD-TBLCA, indicating that concatenation cannot fully exploit different temporal features. BiLSTM generally achieved higher TA-F1 than the transformer, confirming its advantage for precise breakpoint localization. The poorer results of GAM-None and GAM-Original highlight the benefit of enhanced spectral channel attention.

B. Performance Under Sequence Updates

As an ensemble of a transformer and BiLSTM, CCD-TBLCA adopted a maximum temporal length, enabling rapid monthly mapping and change detection without reconfiguration. In Kunming, the 90-month sequences were split

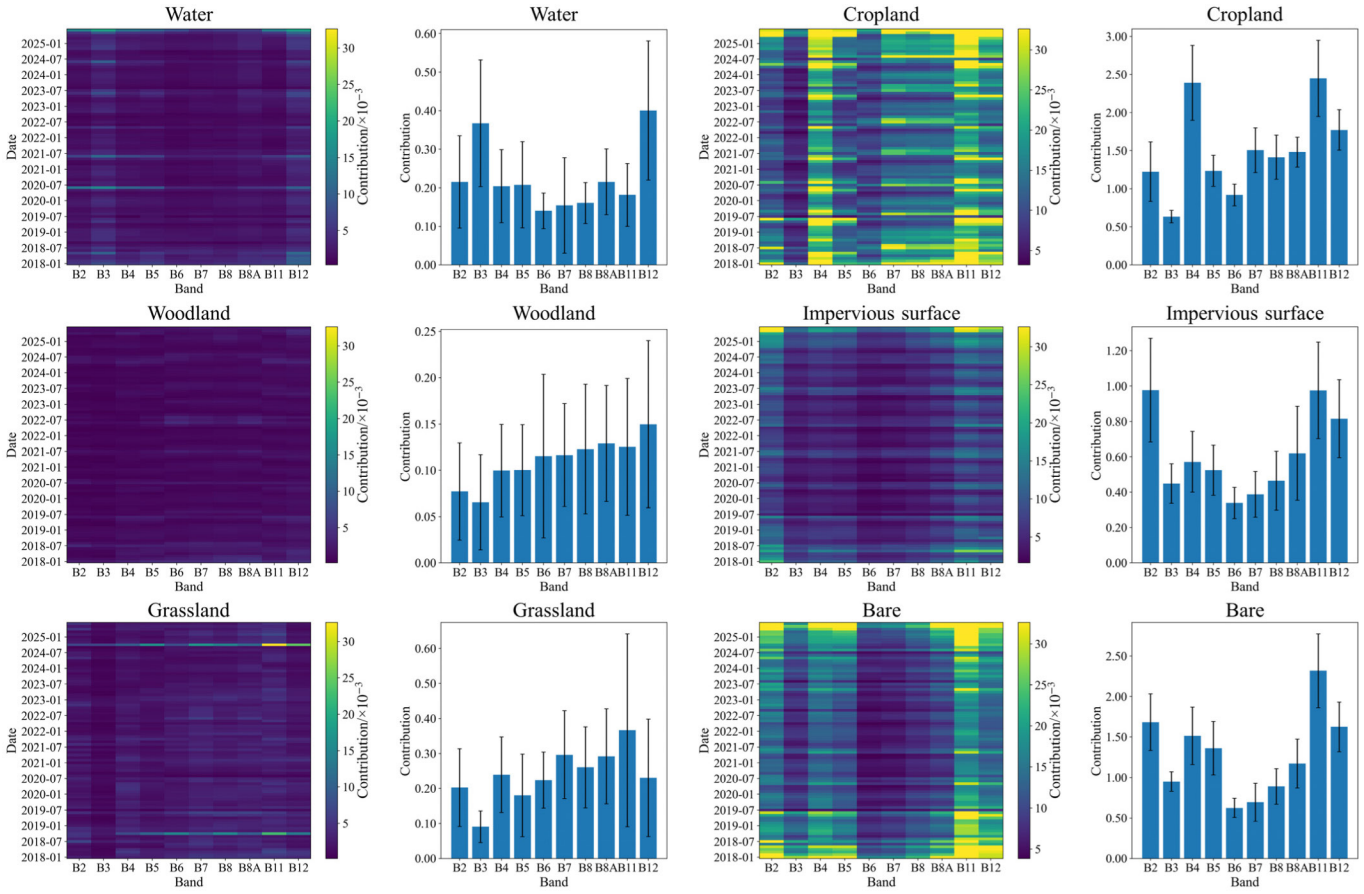


Fig. 25. Spectral-temporal contribution heatmap and importance of different bands for each category.

into January 2018–December 2021 (48 months) and January 2022–June 2025 (42 months). CCD-TBLCA was trained on 48-month historical sequences and tested on the latest sequences. In Fig. 21, Front-48 and Behind-42 denote the detection accuracy within different subsequences, respectively.

Using only historical sequences caused no obvious degradation in temporal classification compared with training on full sequences. For breakpoint detection, relying solely on local sequences reduced SC-F1 by 6.55% and 9.58% in historical and the latest subsequences compared with training on full sequences. This may be because the change patterns in the new data differ from those learned. However, applying the CCD-TBLCA trained on historical sequences directly to the latest sequences did not suffer significant performance loss compared to detecting changes in historical data. As more imagery becomes available, the expanding record is expected to help CCD-TBLCA achieve accurate responses to changes.

C. Robustness to Missing Observations

One design goal was to support direct land cover change detection on irregular time series. CCD-TBLCA used the mask mechanism to exploit available spectral information and achieved high accuracy in the TSSCD-48 dataset and the Kunming dataset. To examine robustness under sparse observations, we applied random masking to simulate a missing proportion from 10% to 60%. As shown in Fig. 22 C-Kappa and C-F1 declined across all cities, with the largest reduction in Xiong'an remaining below 10%. For breakpoint detection in the TSSCD-48 dataset, increasing missingness caused an obvious decline in metric values, whereas Kunming remained relatively stable. In Kunming, even at a 60% missing proportion, CCD-TBLCA maintained accuracy close to that of the original data. In other cities, a 60% missing proportion left only 19 available timesteps, and SC-F1 generally dropped by more than 10%. Overall, the robustness of CCD-TBLCA to missing observations strongly depends on sequence length.

D. Model Interpretation

To assess whether CCD-TBLCA exploited information across the entire sequence, we visually analyzed each MHA in the encoder and decoder. These include the transformer multihead self-attention during encoding, and both the multihead self-attention and multihead CA during decoding. Using the Kunming dataset, we trained CCD-TBLCA ten times and averaged attention weights across heads. Samples were grouped by categories at each timestep. After averaging weights in each group, we performed a weighted average across all training runs using the number of valid records as weights. The resulting weights indicate the relative importance of different timesteps.

Fig. 23 shows the monthly proportion of missing observations and the mean attention weights of the last layer for each MHA. Column-like stripes across the entire sequence indicate that CCD-TBLCA used spectral information at each valid

timestep, while very low weights in months with high missing proportions suggest effective exclusion of clouds and cloud shadows. For the CA using the BiLSTM branch as Query, cropland within the red box in Fig. 23 exhibits clear seasonal dependence with higher weights in key months. Impervious surface and bare land display more temporally stable attention.

Fig. 24 illustrates the role of several attention heads within the final transformer encoding layer. For water, impervious surface, and bare land, which display stable spectral patterns and weak seasonality, attention is nearly uniform across valid timesteps. For woodland, grassland, and cropland with obvious phenology, MHA emphasizes adjacent timesteps and specific nonadjacent months. In particular, the third head exhibits sensitivity to seasonal phase information for woodland and cropland, assigning high weights to timesteps with similar phenological characteristics approximately on an annual cycle.

To analyze the contribution of different bands across months for each category, we employed integrated gradients (IG) to attribute the logit of target type [56] and obtained a spectral-temporal contribution matrix $H_k(t, i)$. $H_k(t, i)$ represents the contribution of band i for the logit of type k at time t. Summing $H_k(t, i)$ over time yielded each band's contribution to type k.

Fig. 25 shows the spectral-temporal contribution heatmap (left) and band contribution (right). For water, classification relies mainly on the green band (B3) and shortwave infrared band (B12), consistent with their reflectance characteristics. Woodland exhibits weak seasonality, consistent with Kunming's predominantly evergreen species. Grassland shows short peaks in red edge, near-infrared, and shortwave infrared bands. Cropland encompasses open-air and greenhouse forms. Bands related to vegetation (red, red edge, near-infrared) and bands sensitive to greenhouse plastic or soil (shortwave infrared) exhibit obvious seasonal fluctuations. Impervious surface is distinguished mainly by the blue band (B2) and shortwave infrared band (B11), potentially due to the spectral reflectance of low-water-content materials like coated steel and concrete. Bare land emphasizes B11, which is sensitive to soil moisture and mineral composition.

VI. Conclusion

This study proposed CCD-TBLCA, a method for detecting continuous land cover changes fusing transformer and BiLSTM time-series encoding information through CA. It captured long-term dependencies across years and seasons while retaining sensitivity to local dynamics, thereby converting spectral time series into continuous land cover information. We also introduced a breakpoint detection criterion that enforced land cover semantic consistency before and after changes, thereby establishing a more comprehensive evaluation framework encompassing spatial existence, temporal accuracy, and semantic consistency.

Experiments on monthly continuous change detection datasets showed that CCD-TBLCA outperformed existing methods such as CCDC and TSSCD in temporal classification and breakpoint detection, with the semantic consistency metric SC-F1 improving by at least 15% in most cities. Compared to existing land cover mapping products, CCD-TBLCA achieved the highest classification accuracy across all land cover types in multiple years, with F1-score for each type generally exceeding 0.9. Ablation experiments confirmed the contributions of channel attention, dual-branch encoding, and CA decoding. Robustness was then examined under incremental sequence updates and varying missing rates. Finally, visualizations of temporal and spectral contributions clarified the decision basis, especially the model's focus on cropland phenology.

In summary, CCD-TBLCA is suitable for continuous land cover change detection in diverse geographical environments. It can accurately represent land cover semantic information before and after changes while maintaining robustness to irregular time series. With its support, monthly tracking of specific land cover types or analysis of change events will be more accurate and reliable. However, a main limitation is the need for large, accurate training samples, which are costly to construct. Future research can focus on extracting high-quality training samples from existing land cover products to support broader model transfer [57], [58].

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